

3. A present is placed at rest on a plane that is inclined, at a distance L from the bottom of the incline that makes an angle θ with the ground. At the bottom of the incline, the box is determined to have a speed v . If the box is instead released from a distance of $4L$ from the bottom of the incline, what will its speed at the bottom of the incline be?
4. Imagine you are swinging a bucket full of water attached to a rope in a vertical circle
- (a) Where along the trajectory is the tension on the rope the highest?
 - (b) Why doesn't the water spill out of the bucket at the top of the circle? Give two explanations:
 - (c) What is the minimum speed required for the water to not spill out of the bucket?